Supplemental Texts (DRAFT)

Pooled-sample genotyping of the ZEAL BC_2S_3 population (zealgt)

Rodríguez-Zapata *et al.*

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1 The crossing scheme and the donor haplotype

Each ZEAL family descends from one teosinte plant crossed to B73. The F_1 receives a single haploid gamete from the teosinte parent, which we call the donor haplotype H_d of donor d. That gamete is a recombinant of the teosinte parent's two haplotypes and exists only in that F_1; it is one sequence, with one allele per site, and needs no phasing. The F_1 is H_d/B73, so every teosinte base pair in its BC_1 progeny and in all of its BC_2S_3 descendants is a piece of H_d. The heterozygosity of the teosinte parent does not enter any model below.

Single-locus expectation. Let h_t be the probability that a plant of generation t is heterozygous H_d/B73 at a locus, and a_t that it is homozygous H_d/H_d. Backcrossing to B73 halves h and creates no homozygotes; selfing halves h and moves the lost half equally into the two homozygous classes:

$$\text{backcross: } h_{t+1} = \tfrac{1}{2}h_t,\ a_{t+1} = 0; \qquad \text{self: } h_{t+1} = \tfrac{1}{2}h_t,\ a_{t+1} = a_t + \tfrac{1}{4}h_t. \tag{1}$$

From $h_{F1} = 1$: $h_{BC1} = \frac{1}{2}$, $h_{BC2} = \frac{1}{4}$, and after s selfings

$$h_{BC2Ss} = \tfrac{1}{4}\,2^{-s}, \qquad a_{BC2Ss} = \tfrac{1}{2}\left(\tfrac{1}{4} - \tfrac{1}{4}\,2^{-s}\right). \tag{2}$$

For BC_2S_3: $h = 1/32 = 3.1\%$, $a = 7/64 = 10.9\%$, B73 $= 55/64 = 85.9\%$, and the donor allele frequency $a + h/2 = 1/8$ is fixed at 12.5% by the two backcrosses, whatever the number of selfings. For BC_2S_2: $h = 1/16$, $a = 3/32$, B73 $= 27/32$. These are the expectations implemented by `single_locus_p0(n_bc, n_self)` (zealhmm, via `nilHMM::breeding_prior`).

2 Pooled samples: BC_1 pools and BC_2S_3 bulks

Every sequenced library in ZEAL is a pool of six plants: each BC_1 sample is six BC_1 plants of one donor, and each BC_2S_3 skim is a six-plant field-plot bulk of one line. A pool of six diploid plants holds twelve chromosome copies at a locus. Whether those twelve copies can be treated as independent draws depends on where each plant's two gametes come from.

2.1 BC_1 pools: six fixed B73 gametes and six F_1 gametes

A BC_1 plant is a B73 gamete times an F_1 gamete. At a site where H_d carries the ALT allele, the B73 gamete is always REF, and the F_1 gamete carries H_d's allele with probability $\frac{1}{2}$, independently across plants. The number of carrier plants in pool i is therefore

$$j_i \sim \text{Binom}(6, \tfrac{1}{2}), \qquad f_i = \frac{j_i}{12} \in \{0, \tfrac{1}{12}, \ldots, \tfrac{6}{12}\}, \tag{3}$$

with $E[f_i] = \frac{1}{4}$, $\text{SD}[f_i] = \sqrt{6 \cdot \frac{1}{4}}/12 = 0.102$, and $f_i \le \frac{1}{2}$ always. Treating the twelve copies as independent haplotypes, each ALT with probability $\frac{1}{4}$ (Binom$(12, \frac{1}{4})$), would keep the mean but inflate the variance ($12 \cdot \frac{1}{4} \cdot \frac{3}{4} = 2.25$ vs. 1.5 haplotypes) and allow $f > \frac{1}{2}$, which the cross cannot produce. The plant level is what matters here, because the two gametes of a plant have different sources. This is the model of step 4 of discovery (Text 4).

Presence. A segment of H_d is absent from all $6k$ F_1 gametes of a donor's k samples with probability 2^{-6k}, so it is present in the pooled DNA with probability $1 - 2^{-6k}$ (98.4% for one sample, $> 99.99\%$ for three). Completeness of H_d is limited by detection (Text 3), not presence. Along the chromosome j_i is constant between the crossovers of the six F_1 gametes, so neighbouring sites share j_i; the per-site model of Eq. (3) ignores that correlation.

2.2 BC_2S_3 bulks: sibs of one selfed parent

The six plants of a BC_2S_3 plot are sibs from selfing one BC_2S_2 parent. At a locus the parent is B73/B73, H_d/H_d, or heterozygous, with the BC_2S_2 frequencies of Text 1. If it is homozygous the bulk's donor fraction is fixed at 0 or 1. If it is heterozygous, every sib gamete carries H_d with probability $\frac{1}{2}$ independently, so each sib's dosage is $\mathrm{Binom}(2, \frac{1}{2})$ and the bulk's donor-haplotype count is a sum of six of them:

$$K \sim \mathrm{Binom}(12, \tfrac{1}{2}), \qquad g = \frac{K}{12} \in \{0, \ldots, 1\}, \qquad E[g] = \tfrac{1}{2},\ \mathrm{SD}[g] = 0.144. \tag{4}$$

Here, unlike the BC_1 pool, grouping the twelve gametes into plants does not change the distribution at a single locus: both gametes of a sib come from the same parent. The marginal donor fraction of a bulk is the mixture

$$P(g) = \tfrac{27}{32}\,\delta_0 + \tfrac{3}{32}\,\delta_1 + \tfrac{1}{16}\,\mathrm{Binom}\big(12g; 12, \tfrac{1}{2}\big)\,, \qquad E[g] = \tfrac{1}{8}, \tag{5}$$

which keeps the 12.5% donor content of Text 1. The bulk's parent is BC_2S_2. The chr10 genotype frequencies of the two full-coverage mexicana families (84 lines) are closer to the BC_2S_2 expectation, with an excess of HET and a deficit of introgression against it (PLAN_pipeline §4 #12). Within a segment where the parent is heterozygous, K is constant between the breakpoints of the twelve gametes, so a segment's read fraction is one draw of $K/12$ (e.g. 3/12 or 10/12), not $\frac{1}{2}$. A single-plant three-state model with emission means $\{\epsilon, \frac{1}{2}, 1-\epsilon\}$ mis-specifies such segments; the alternative is a state space on the parent's genotype (a BC_2S_2 prior) with Eq. (4) as the heterozygous emission. Which fits the ZEAL bulks better is an empirical question (Text 6).

2.3 Unequal DNA per plant

If plant p contributes a DNA share w_p ($\sum_p w_p = 1$), a read comes from plant p with probability w_p, and both haplotypes of that plant share its weight. The expected ALT fraction becomes $\sum_p w_p\, x_p/2$ ($x_p \in \{0, 1, 2\}$ the plant's dosage) instead of the count-over-12 of Eqs. (3)–(4). With $w_p = \frac{1}{6}$ the equal-pooling models are exact; unequal w_p spreads the fractions off the $1/12$ lattice and adds overdispersion shared by the two haplotypes of a plant. **[TO CHECK: measure pooling evenness from the sharpness of the $j/12$ lattice in deep BC_1 samples]**

3 Reads at low coverage

With mean coverage λ per sample, the reads at a site are Pois(λ) (Lander–Waterman). At the BC_2S_3 skim depths:

λ	0 reads	1 read	2 reads	≥ 3 reads
0.25×	0.78	0.19	0.024	0.002
0.4×	0.67	0.27	0.054	0.008
0.67×	0.51	0.34	0.11	0.03
1.2×	0.30	0.36	0.22	0.12

At 0.4× a covered site almost always has one read, which samples one of the bulk's twelve copies; no single site separates heterozygous from homozygous, and dosage can only come from aggregating reads along a segment (the ancestry HMM of Text 6). The ALT reads at a site thin the Poisson: if the sample's ALT fraction is f, the ALT reads are Pois(λf) and the probability of seeing none is $e^{-\lambda f}$ (used in Eq. (8)).

Duplicates. Every model here treats reads as independent draws from the pool. A PCR duplicate is the same molecule counted twice, so it inflates both n and the ALT count a without adding information, and the likelihood ratios of Text 4, which are sums over reads, overstate the evidence by roughly the duplication factor. At skim depth this is negligible; in the BC_1 samples (5–25× each), where one carrier plant contributes about $\lambda/12$ ALT reads, it can push weak sites over the thresholds. Duplicate removal before variant calling is standard in pooled-sequencing pipelines (Huang et al., 2015) and in GATK best practices (Van der Auwera et al., 2013); the CRISP paper does not discuss it (Bansal, 2010).

4 Per-donor variant discovery

Candidate sites come from CRISP (Bansal, 2010) run on the donor's BC_1 samples plus a witness pool (the donor's BC_2S_3 lines merged). Our post-filter (step 4) scores each candidate per donor.

4.1 Pooled likelihood ratio

For pool i of donor d with n_i reads, a_i of them ALT, and a site error rate ϵ,

$$L_1^{(i)} = \sum_{j=0}^{6} \binom{6}{j} 2^{-6} \,\mathrm{Binom}\big(a_i;\, n_i,\, p_j\big), \qquad p_j = \tfrac{j}{12}(1-\epsilon) + \big(1-\tfrac{j}{12}\big)\epsilon, \tag{6}$$

$$L_0^{(i)} = \mathrm{Binom}(a_i;\, n_i,\, \epsilon), \qquad \mathrm{LLR}_d = \sum_{i \in d} \left(\log L_1^{(i)} - \log L_0^{(i)}\right). \tag{7}$$

j_i is marginalised per pool because it is not observed (Eq. (3)); $j_i = 0$ is allowed, so a pool with no ALT reads is not evidence against H_d's allele. With prior π the posterior is $\sigma(\mathrm{LLR}_d + \log \frac{\pi}{1-\pi})$; at $\pi = \frac{1}{2}$ the tier thresholds $\mathrm{LLR} \geq 6.9,\ 4.6,\ 2.2$ are posteriors $\geq 0.999,\ 0.99,\ 0.9$. A site is REF (tier `ref`) when $\mathrm{LLR} \leq -4$ with pooled depth ≥ 12.

Site error rate. ϵ is estimated per site from the pools of donors that are confidently non-carriers there ($\mathrm{LLR} < -3$ at the starting value), including two B73 pools counted at the candidate sites, when those hold ≥ 20 reads; otherwise a default is used.

Flags. `af_gt_half`: the donor's pooled ALT fraction is significantly above $\frac{1}{2}$ (binomial test), impossible under Eq. (3) and the signature of paralogs or mis-mapping. `hidepth`: summed depth $> 2\times$ the median, the signature of copy-number gain. Tier A additionally requires ALT reads in ≥ 2 BC_1 samples.

4.2 Witness veto

The veto keeps a candidate only if the witness pool shows ≥ 1 ALT read. For a real H_d allele, a line of coverage λ shows no ALT read with probability $E[e^{-\lambda g}]$ over the bulk fraction g of Eq. (5); using $E[e^{-\lambda K/12}] = \big(\frac{1+e^{-\lambda/12}}{2}\big)^{12}$,

$$q(\lambda) = \tfrac{27}{32} + \tfrac{3}{32}e^{-\lambda} + \tfrac{1}{16}\left(\tfrac{1+e^{-\lambda/12}}{2}\right)^{12}, \qquad P(\text{real allele vetoed}) \approx q(\lambda)^L \tag{8}$$

for L lines, treating lines as independent at the locus (they are not: sibling lines share BC_1 and BC_2 ancestry, which makes the true loss larger). $q(0.4) = 0.958$ and $q(1.2) = 0.907$, so the veto

discards about 65% of real alleles with ten lines at 0.4× (0.958^{10}), 19% with 38 lines, and 25% with fourteen lines at 1.2×. This is consistent in direction with the pilot, where a 10-line witness kept 20% of CRISP records and a 38-line witness 51% (chr10; records include artifacts, so the numbers are not directly comparable). **[TO CHECK: compare Eq. (8) with the QC-set witness]**

5 The union of informative sites and gap filling

5.1 Genotype as ancestry times donor allele

For a line of donor d at site s, with ancestry dosage $x \in \{0, 1, 2\}$ (copies of H_d's segment) and donor allele $D_{ds} \in \{\text{REF}, \text{ALT}\}$, the genotype is fixed:

$$G_s = \begin{cases} x & D_{ds} = \text{ALT} \\ 0 & D_{ds} = \text{REF} \\ \text{unknown} & D_{ds} \text{ missing and } x > 0. \end{cases} \tag{9}$$

A common marker set across donors therefore needs D_{ds} at every site of the union $U = \bigcup_d A_d$ of the donors' tier-A sets, including sites discovered only in other donors. Positions with different ALT alleles in different donors are dropped.

Terminology: informative sites and false alleles. Whether a site is informative is a property of the donor: site s is *informative* for donor d if $D_{ds} = \text{ALT}$, and *non-informative* if $D_{ds} = \text{REF}$ (the cross is not polymorphic there). A *false allele* or a *missed allele* is an error of the call. Crossing the truth with the call:

	$D_{ds} = \text{REF}$ (non-informative)	$D_{ds} = \text{ALT}$ (informative)
called ALT	false allele	true allele
called REF	correct REF	missed allele (false REF)

Every non-informative site is in the left column, whatever its call; a false allele is a non-informative site called ALT. In the simulated QC set (founder inbred, no standing variation) the two coincide, which is why zealbc1 used "non-informative" for the false alleles of pooled discovery; in real donors they are distinct. One source of false alleles is donor-only sequence: reads from a donor copy absent from B73 align to a B73 paralog at s, where the donor is REF. Such reads can align with high mapping quality, so a MAPQ filter does not remove them, and they appear only in carriers of the donor segment that holds the copy.

5.2 Gap filling

The donor's BC_1 samples are counted at every site of U (one pileup per sample, all union sites at once), its own sites ($s \in A_d$) included, and scored by Eq. (7) without the veto. REF is assigned from the donor's own reads only (tier `ref`). ALT uses a prior raised by the other donors: with $k =$ other donors that carry the allele (s among their own sites) and $m =$ other donors with a genotype at s (own site, or a gap scored `ref`),

$$\pi_{ds} = \frac{w\mu_d + k}{w + m}, \qquad P(D_{ds} = \text{ALT} \mid \text{reads}) = \sigma\left(\text{LLR}_{ds} + \log\frac{\pi_{ds}}{1 - \pi_{ds}}\right), \tag{10}$$

a leave-one-out beta-binomial prior with mean μ_d (donor d's sharing rate: tier A/(A+ref) at its gaps) and weight $w = 2$. ALT is called at posterior ≥ 0.999 with reads at s; sites flagged `hidepth`

or `af_gt_half` are never promoted. Own sites are re-scored the same way but are never REF: an own site whose posterior falls below 0.999 becomes missing, also when its reads at the union are tier `ref` (then discovery and the second count disagree). The prior comes from all other donors; restricting it to the same taxon is an option once other taxa are added. The likelihood ratio models sequencing error but not artifacts; a site that is tier A in another donor, passed that donor's witness and is clean in the B73 pools is independent evidence that the site is real, which is what the prior carries. A mis-mapped paralog shows ALT in all donors, so sharing alone does not separate real alleles from artifacts; the B73 zero class and the flags do.

Expected prior from the teosinte panel. μ_d and the k, m of Eq. (10) are estimated within a run, from the donors' own calls, so no external prior is needed. Their expected values come from a genotyped panel, without assemblies. For individual i at site s, $q_i(s) = G_i(s)/2$ is the probability that its gamete carries ALT ($G_i \in \{0, 1, 2\}$ ALT copies). For a donor of taxon B given a donor of taxon A that carries the allele,

$$c_{AB} = \frac{\sum_{i\in A}\sum_{j\in B,\, j\neq i}\sum_s q_i(s)\, q_j(s)}{\sum_{i\in A}\sum_{j\in B,\, j\neq i}\sum_s q_i(s)\,\mathbb{1}[j \text{ called at } s]} \;\approx\; \frac{E_s[p_A(s)\, p_B(s)]}{E_s[p_A(s)]}, \tag{11}$$

the probability that j's gamete carries ALT given that i's does, with $p_A(s)$ the ALT frequency in taxon A. It is the quantity μ_d estimates (with B the donor's taxon and A the taxa of the other donors). The unconditional rate is $d_B = E_s[p_B(s)]$, the transmitted donor dose. Within a taxon $c_{BB} = E[p_B^2]/E[p_B] \geq E[p_B] = d_B$, so the prior mean of Eq. (10) lies above d_B even when nothing is wrong. On the teosinte panel of Grzybowski et al. (2023) (217 individuals, 49,002 SNP50K sites, biallelic, minor allele frequency ≥ 0.05; "other" = diploperennis, luxurians and huehuetenangensis, the cluster these donors form in the ZEAL PCA):

donor taxon B	n	d_B	$c_{\text{mexicana},B}$	$c_{\text{parviglumis},B}$	$c_{\text{other},B}$
mexicana	79	0.312	0.662	0.567	0.453
parviglumis	88	0.264	0.482	0.551	0.401
other	15	0.332	0.485	0.504	0.677

Sharing within a group is about twice d_B, and the "other" group shares most within itself (0.68; luxurians, diploperennis and nicaraguensis each 0.64–0.70 with it), which supports taking the prior donors of Eq. (10) from the donor's group. The panel sites are common variants, while tier-A sites also include rare and private alleles, so these values are upper bounds for μ_d: the two mexicana pilot donors gave $\mu_d = 0.52$ and 0.44 (chr10:1–20 Mb), between d and $c_{\text{mexicana,mexicana}}$. The simulated QC set (inbred assembly founders Gigi and TIL18, all assembly SNPs) gave $\mu_d = 0.24$ and 0.20, below both; it does not model outbred donors. A run's μ_d outside $[d_B, c_{BB}]$ flags the donor.

5.3 Gap filling, step 2: evidence from the lines

Equation (10) leaves a gap missing when the BC_1 reads are neither REF (tier `ref`) nor ALT at posterior ≥ 0.999. The donor's BC_2S_3 lines add independent evidence, because the ancestry inference (Text 6) tells, for every line i and site s, how many copies of H_d's segment the line carries, $x_i \in \{0, 1, 2\}$. A read from a donor copy shows D_{ds}; a read from a B73 copy shows REF whatever D_{ds} is. With n_i reads and a_i ALT reads in line i, and c_s the mappability of the donor copy relative to B73 at s (the share of donor-copy reads that reach s; the teosinte copy maps less well where it

has diverged),

$$n_i \sim \mathrm{Pois}\big(k_s\lambda_i\,[(1-\tfrac{x_i}{2})+\tfrac{x_i}{2}c_s]\big), \tag{12}$$

$$a_i \mid n_i,\ D_{ds}=\mathrm{ALT} \sim \mathrm{Binom}\big(n_i,\ \rho_i(1-\epsilon)+(1-\rho_i)\epsilon\big), \qquad \rho_i=\frac{\frac{x_i}{2}c_s}{(1-\frac{x_i}{2})+\frac{x_i}{2}c_s}, \tag{13}$$

$$a_i \mid n_i,\ D_{ds}=\mathrm{REF} \sim \mathrm{Binom}(n_i,\ \epsilon), \tag{14}$$

with λ_i the line's coverage (from its B73 segments), k_s the site's B73 mappability (from the B73-ancestry lines at s), and c_s summed out under a per-taxon prior (mexicana: from simulated TIL18 reads against B73 reads, median 0.85, 10.6% of sites below 0.25). Lines with $x_i = 0$ contribute the same term under both hypotheses and cancel. The evidence of the lines, $\mathrm{LLR}_{ds}^{\mathrm{lines}} = \log L(\mathrm{ALT}) - \log L(\mathrm{REF})$, is independent of the BC_1 reads given D_{ds}, so the log-odds add:

$$P(D_{ds}=\mathrm{ALT} \mid \mathrm{BC}_1,\ \mathrm{lines}) = \sigma\left(\log\frac{\pi_{ds}}{1-\pi_{ds}} + \mathrm{LLR}_{ds} + \mathrm{LLR}_{ds}^{\mathrm{lines}}\right), \tag{15}$$

and ALT is called at the same cut-off as Eq. (10), posterior ≥ 0.999. Step 2 never calls REF: one ALT read in a line with $x_i = 2$ is $\sim 1/\epsilon \approx 200$ times more likely under ALT, while ruling ALT out needs many donor-copy reads, and at the sites that step 1 leaves missing there are about 0–1 (≈ 5 donor-segment lines among 40 at $\sim 0.9\times$, the donor copy mapping poorly).

On the simulated QC set (TIL18; 10 lines at 0.8× and 1.2×; true and inferred ancestry; truth = the founder's own reads, see below) Eq. (15) made **no false ALT call at any of the 29,098 truly-REF gap sites**, also with LLR_{ds} set to 0 (false-ALT rate $< 10^{-4}$, 95% upper bound), and recovered 29–46% of the 11,918 truly-ALT gap sites from the lines and prior alone and 93–94% with the BC_1 evidence. On the two mexicana pilot donors (chr10) it filled 2,994 and 2,581 of the 11,819 and 9,079 gaps left by step 1, leaving 10.7% and 7.9% of the union missing.

5.4 Gap filling, step 2 as joint genotyping

[PROPOSAL, not adopted (2026-10-05)]

Eq. (15) conditions on one ancestry call per line. The donor allele D_{ds} is a parameter shared by all of donor d's samples at site s; each line's ancestry at s is a latent variable. Summing the lines' likelihoods over their ancestry, instead of plugging in a call, makes step 2 the analogue of joint genotyping in GATK (Van der Auwera et al., 2013): a population prior (the other donors, Eq. (10)), per-sample likelihoods with their latent states marginalised, and one posterior per site. It stays a single pass: ancestry is inferred once, from the donor's own tier-A sites, and nothing is fed back.

Latent state of a line. Line i is a six-plant bulk (Text 2.2). At site s its $\mathrm{BC}_2\mathrm{S}_2$ parent is $z \in \{\mathrm{B73/B73},\ \mathrm{HET},\ H_d/H_d\}$, and the bulk's donor fraction is

$$g=0\ \ (z=\mathrm{B73/B73}), \qquad g=\tfrac{K}{12},\ K\sim\mathrm{Binom}(12,\tfrac{1}{2})\ \ (z=\mathrm{HET}), \qquad g=1\ \ (z=H_d/H_d), \tag{16}$$

by Eq. (4). This replaces the dosage $x_i/2$ of Eq. (14), which is the single-plant value and the mean of $K/12$.

Emission. With the mappabilities k_s, c_s and coverage λ_i of Eq. (14),

$$n_i \mid g \sim \mathrm{Pois}\big(k_s\lambda_i\,[(1-g)+g\,c_s]\big), \qquad \rho(g)=\frac{g\,c_s}{(1-g)+g\,c_s},$$
$$a_i \mid n_i, g,\ D_{ds}=\mathrm{ALT} \sim \mathrm{Binom}\big(n_i,\ \rho(g)(1-\epsilon)+(1-\rho(g))\epsilon\big), \qquad a_i \mid n_i, g,\ D_{ds}=\mathrm{REF} \sim \mathrm{Binom}(n_i,\ \epsilon). \tag{17}$$

Ancestry posterior. Let $w_i(z) = P\big(z_i(s) = z \mid \mathbf{y}_{i,-s}\big)$ be line i's posterior at s given its reads at every site except s. A gap site is not among the donor's tier-A sites, so the ancestry HMM never saw it and w_i is its forward–backward posterior at an unobserved position: with $l < s < r$ the flanking markers, α, β the forward and backward variables, and $T(\delta)$ the transition matrix over distance δ,

$$w_i(z) \;\propto\; \Big[\sum_{z'} \alpha_{il}(z')\, T_{z'z}(s-l)\Big]\Big[\sum_{z''} T_{zz''}(r-s)\, \beta_{ir}(z'')\Big]. \tag{18}$$

Joint posterior. Given D_{ds} the lines are independent, and independent of the BC_1 pools, so

$$L_i(D) = \sum_z w_i(z) \sum_K P(K \mid z)\; P\big(n_i, a_i \mid g(z,K),\, D\big), \tag{19}$$

$$P(D_{ds} = \text{ALT} \mid \text{all reads}) = \sigma\Big(\log\frac{\pi_{ds}}{1-\pi_{ds}} + \text{LLR}_{ds} + \sum_i \log\frac{L_i(\text{ALT})}{L_i(\text{REF})}\Big), \tag{20}$$

with π_{ds} from Eq. (10), LLR_{ds} from the BC_1 pools (Eq. (7), which already sums over each pool's carrier count j), $P(K \mid \text{HET}) = \text{Binom}(K; 12, \frac{1}{2})$ and K fixed for the homozygous states. c_s is summed out under the per-taxon prior of step 2. A line with $w_i(\text{B73/B73}) = 1$ gives $L_i(\text{ALT}) = L_i(\text{REF})$ and drops out. ALT is called at posterior ≥ 0.999, as in Eq. (15), which Eq. (20) reduces to when every w_i is a point mass and $K/12$ is replaced by $\frac{1}{2}$.

Why it is not circular. The reads at s enter Eq. (20) once: w_i uses only the line's other sites. For an own site ($s \in A_d$, already in the HMM) the same holds with the site's own emission removed, $w_i(z) \propto \alpha_{is}(z)\,\beta_{is}(z)/e_{is}(z)$; whether own sites are re-scored with the lines is a separate choice.

Not proposed. A fully simultaneous model, with $D_{d\cdot}$ at every site, each BC_1 pool's carrier count $j(s)$ and each line's parent state $z(s)$ as hidden Markov chains along the chromosome, couples all chains through D and needs EM or Gibbs sampling, with an own HMM in place of RTIGER. Eq. (20) keeps RTIGER and one pass.

Open points.

- **[TO CHECK: does RTIGER export per-site posterior probabilities and its fitted transition parameters? If not, Eq. (18) needs a forward–backward pass on RTIGER's fitted model]**
- **[TO CHECK: RTIGER's states are single-plant dosages; mapping them to the parent states z of a bulk]**
- **[TO CHECK: lines of one donor share BC_1 and BC_2 ancestry, so they are not independent given D_{ds} (as for the veto, Eq. (8)); the product over lines overstates the evidence]**
- **[TO CHECK: benchmark against Eq. (15) on the simulated QC set: false-ALT rate at truly-REF gaps and recovery of truly-ALT gaps]**

[end of proposal]

5.5 Benchmark

On the simulated QC set (founders Gigi and TIL18, five BC_1 pools of $\sim 14\times$, tier A subsampled to the pilot's sizes; chr10) the first truth was the founder assembly's AnchorWave SNPs in the lowcopy ranges, with every other position taken as REF. Against it, gap ALT at posterior ≥ 0.999 was truly ALT at the rate of the donors' own tier-A sites (0.74 and 0.75 vs. 0.68 and 0.74), gap REF was 97–98% correct, and the Bayes rule and the heuristic "tier A or B" rule gave the same precision at this depth. At most of the sites scored as false, however, reads simulated from the same assembly and aligned with minibwa show the non-B73 base at normal depth: the assembly alignment and the read alignment place that sequence differently (or the position is unaligned, or the SNP was lost in the extraction). Because every step of the pipeline works on minibwa-aligned reads, the benchmark now uses as truth what the founder's own reads show at each site (ALT if $\geq 50\%$ of ≥ 5 reads are ALT, REF if none of ≥ 5 reads is, unknown otherwise). Against it the precision is ≈ 0.97 for own tier-A sites and ≈ 0.99 for gap ALT. This is the right reference for the read-based steps, not a better truth about homology.

5.6 Size of the union

The union U sets the cost of the union counts, of gap filling and of every per-line table, so its growth with the number of donors n is projected before a run over all donors. The data are the sharing patterns of the four-donor chr10 union (four mexicana donors: which donors' tier-A sets contain each union site; notebook `docs/notebooks/03_union_rarefaction.qmd`).

Starting point: the allele frequency spectrum. Let a site's ALT frequency among donors be p, with the neutral spectrum $g(p) \propto 1/p$; each donor carries the ALT with probability p, independently, and finds a carried ALT with probability η (discovery sensitivity). A site is missing from the union of n donors only if no donor finds it, so over S sites

$$E[|U_n|] = S \sum_p g(p)\,[1 - (1 - \eta p)^n]\,. \tag{21}$$

With one η for every donor, Eq. (21) predicts 1.74–1.95 times the two-donor union at four donors ($\eta = 0.1$–0.3); the test gave 2.43 (70,210 to 170,729 sites), because the two added donors each find twice as many sites as the first two. The observed curve is therefore fitted directly. Its points are the *rarefaction curve*, the mean union over all $\binom{4}{n}$ donor sets of size n: $\bar{U}_1, \ldots, \bar{U}_4 =$ 62,149, 106,965, 141,993, 170,729.

Michaelis–Menten: an empirical curve from ecology. Species accumulation curves are routinely fitted with

$$\bar{U}_n = \frac{U_\infty\, n}{K + n}, \tag{22}$$

two parameters: the asymptote U_∞ and the number of donors K that reach half of it (Soberón and Llorente, 1993). It is interpretable as follows. If site s is found by each donor independently with rate k_s, it is in the union of n donors with probability $1 - e^{-k_s n}$; with k_s gamma-distributed across sites (shape α, scale $1/K$),

$$\bar{U}_n = U_\infty \left[1 - \left(1 + \frac{n}{K}\right)^{-\alpha}\right], \tag{23}$$

which is Eq. (22) at $\alpha = 1$ and the negative exponential $U_\infty(1 - e^{-kn})$ as $\alpha \to \infty$. Since $k_s = -\log(1 - \eta p_s) \approx \eta p_s$, α is the shape of the spectrum near $p = 0$: the neutral spectrum $1/p$ is

the limit $\alpha \to 0$, which gives logarithmic growth without an asymptote (Fisher's log-series, Fisher et al., 1943). Michaelis–Menten, $\alpha = 1$, is a spectrum with fewer rare alleles than neutral and a finite pool of sites. Fitted to the four points: $U_\infty = 414{,}000$ and $K = 5.7$ (RMSE 418 sites), 263,000 sites at ten donors.

Watterson with false alleles. If B73 and the n donors were $n+1$ haplotypes of one neutral population, the union would be the segregating sites of that sample, with expectation $\theta \sum_{i=1}^{n} 1/i = \theta H_n$ (Watterson, 1975). Two additions make it fit the data: the sites where every mexicana differs from B73 (fixed differences, c) are in the union from the first donor, and a false allele (Text 5) is private to the donor that called it, so false alleles add a constant number ε_f per donor (f for false; ϵ alone is the sequencing error rate):

$$\bar{U}_n = c + \theta_\eta H_n + \varepsilon_f \, n, \tag{24}$$

with θ_η the θ of the sites found. Fitted: $\theta_\eta = 63{,}831$, $\varepsilon_f = 13{,}204$ per donor, against $(1 - 0.74) \times 62{,}149 = 16{,}159$ expected from TIL18's tier-A precision on the QC set (0.74, Benchmark above) if every false allele were private; false alleles from a donor-only copy carried by several donors are shared and fall into $\theta_\eta H_n$ instead. $c = -14{,}947$ is negative, which the model does not allow; three parameters on four points leave one degree of freedom, so this is a consistency check. Ten donors: 304,000 sites.

Use. Both curves are fitted to four mexicana donors and agree near the data (263 and 304 thousand sites at ten donors). Far from it they differ in kind: Eq. (22) stops at U_∞, Eq. (24) grows without limit, and the false alleles alone grow linearly (≈ 1.25 M on chr10 at 95 donors at the fitted ε_f). Donors of other taxa share fewer sites than these four.

6 Ancestry inference, genotype imputation, and their checks

Ancestry inference. Per donor and chromosome, RTIGER (Campos-Martin et al., 2023) (nilHMM's port) on the donor's own tier-A sites, with the REF and donor-ALT read counts of each line; RTIGER learns its transition and emission parameters from the data, so no design prior enters. Rigidity $r = \max(1, \mathrm{round}(0.005\, n_d))$, with n_d the donor's tier-A sites on the chromosome (an Arabidopsis 250 per ∼50 k SNPs carried over as a ratio). A line with fewer than $2r$ covered markers (at least one read) on the chromosome is excluded before inference: RTIGER stops below that. The states are RTIGER's single-plant dosages $x \in \{0, 1, 2\}$, read as they are although each line is a bulk (Text 2.2). Inference gives segments, from a run's first marker to its last.

Grid. The union sites, placed on the B73 v5 consensus map (a monotone spline through its bp–cM pairs), thinned to markers at least 0.1 cM apart by the greedy sweep (sort by cM, keep the first, skip all within 0.1 cM; exact on a line, Jena et al., 2018); one grid per chromosome, shared by all donors. Each line's segments are read at the grid; a marker between two segments is missing. These feed the QTL scan (R/qtl `bcsft`, BC_2S_2, Haley–Knott), one population per donor, and a VCF whose genotype is the ancestry dosage on the union site's REF and ALT.

Genotype imputation. Per donor, a two-founder PHG graph (B73 and the donor founder built from $D_{d\cdot}$ on U), imputing only that donor's lines, rasterised at U (Eq. (9)). These feed GWAS.

Checks. (i) Per-line Mb fractions REF/HET/ALT against the single-locus expectation (Text 1; single plant) or the bulk mixture (Eq. (5)), with a Hotelling test on the (HET, ALT) fractions; the expectation for the bulks is that of their BC_2S_2 parent. (ii) B73 controls free of donor segments. (iii) Segment-length distribution against the expectation for bulks of a BC_2S_2 parent. [**TO CHECK: derive the expected segment-length distribution of a bulk from Eq.** (4)]

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